



✓ EXPERIMENT NO. 7

SPEED CONTROL OF INDUCTION MOTOR BY VARIABLE FREQUENCY METHOD

AIM

To control the speed of the 3 phase induction motor by changing the supply frequency and to plot the speed Vs frequency curve.

APPARATUS

Sl. No.	Name of the apparatus	Type	Range	Quantity
1.	Voltmeter			
2.	Frequency meter			
3.	Ammeter			
4.	Ammeter			
5.	Tachometer			
6.	Rheostat			
7.	Rheostat			

PRINCIPLE

The synchronous speed N_s of an induction motor is related to supply frequency f and number of poles P by the equation, $N_s = \frac{120f}{P}$. Rotor speed is given by $N = (1-s)N_s$ where s is the slip. The basic methods of speed control of an induction motor are a) by changing the number of poles and b) by varying the line frequency.

The emf per phase of an induction motor is given by $E = 4.44K_w f \Phi T$ volts. The induced emf E is nearly equal to the applied voltage V (neglecting drop in stator V impedance). Thus, we can write $\frac{V}{f} = 4.44K_w \Phi T$

When the frequency is reduced, the applied voltage also must be reduced proportionally so as to maintain constant flux, otherwise the core will get saturated resulting in excessive core loss and magnetizing current. The maximum torque also remains constant under this condition. However, the voltage is not varied proportionately in the low frequency range to account for the voltage drop in the winding resistance. This type of control (constant V/f) is used for speed control below base frequency (line frequency of 50Hz).